

Active Directory, PowerShell & RBAC

Project type: Hands-on identity administration lab Environment: Windows Server 2022 / VMware Player

Project overview

Used Windows PowerShell and Active Directory Domain Services to inspect lab user identities, validate security-group assignments, and review group properties relevant to role-based access control (RBAC). The exercise emphasizes verification and least-privilege reasoning rather than claiming production administration experience.

Lab environment

Component	Configuration
Domain controller	JMC-DC01
Active Directory domain	jmc-local
Platform	Windows Server 2022, VMware Workstation Player
Tools	PowerShell, Active Directory module

Objectives

- Identify test accounts and confirm their logon names.
- Inspect membership of HelpDesk and IT Support security groups.
- Review the scope and category of a privileged IT group.
- Connect group assignments to RBAC and least-privilege principles.

Verified lab observations

Accounts queried: Sarah Miller (smiller), Emily Carter (ecarter), Michael Reed (mreed).

Membership mapping: GG-IT-Support included Emily Carter and Michael Reed; GG-IT-HelpDesk included David Williams (dwilliams).

Group properties: GG-Privileged-IT-Admins was reviewed as a Global / Security group. These observations describe a controlled lab, not a live enterprise environment.

Security rationale

Security groups allow access to be assigned by job function rather than through scattered individual permissions. Membership review helps identify excessive or unexpected access. A privileged group warrants additional scrutiny because its membership can increase the impact of compromised credentials.

PowerShell verification examples

The commands below illustrate the type of checks used. They are provided as reproducible examples; exact command history should be matched to original screenshots before publication.

```
Get-ADUser -Filter * | Select-Object Name, SamAccountName  
Get-ADGroupMember -Identity "GG-IT-Support"  
Get-ADGroupMember -Identity "GG-IT-HelpDesk"  
Get-ADGroup -Identity "GG-Privileged-IT-Admins" -Properties GroupScope,GroupCategory
```

Outcome

Validated sample Active Directory identities, reviewed IT security-group membership, and examined privileged-group properties. This demonstrates foundational IAM auditing, PowerShell-based directory queries, and RBAC verification.

Portfolio evidence status

Screenshots not embedded yet. Original Lab 5 screenshots were not available for verification when this report was prepared. Recommended evidence: (1) AD user query results, (2) HelpDesk and IT Support group-membership output, and (3) privileged-group scope/category output. Add only clear, redacted screenshots that match the findings above.

Skills demonstrated

Active Directory Domain Services • PowerShell • Identity auditing • Group membership verification • RBAC • Least privilege • Windows Server 2022

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